

Chapter 1

Introduction

Project Context

The jeepney, a symbol of Philippine resilience and ingenuity, has a rich history. It originated from surplus U.S. military jeeps repurposed after World War II, quickly becoming the backbone of the nation's public transportation system (Lico, 2012). While initially utilitarian, jeepneys evolved into vibrantly decorated icons, showcasing the culture and artistry of their owners. Over time, they expanded beyond urban centers to serve remote rural communities, solidifying their role in the Philippine transportation landscape (De Guzman, 2018).

In Davao City, as in much of the Philippines, public transportation remains a vital part of daily life, with jeepneys serving as a primary mode of transport. However, this traditional system faces significant challenges. Inconsistent fare collection, unpredictable passenger flow, and limited accessibility for persons with disabilities (PWDs) create daily difficulties for commuters and financial instability for jeepney drivers (DOTr, 2017).

Davao City, with an estimated population of 2.03 million in 2025, sees approximately 60% of its residents relying on public transportation, underscoring the critical need for a reliable system (PSA, 2020). The ongoing Public Utility Vehicle

Modernization Program (PUVMP), mandated by the Department of Transportation, aims to replace traditional jeepneys with modern, compliant units. However, this has placed a considerable financial burden on drivers, who struggle with financing and meeting the program's requirements (LTFRB, 2017; David & Cruz, 2021).

Commuters frequently experience long waits and difficulty finding rides, particularly during rush hours. Additionally, PWDs face obstacles due to the lack of accessible features. Research emphasizes commuter struggles in Davao City, highlighting issues such as traffic congestion and inadequate infrastructure, further underscoring the need for innovative solutions (Reyes et al., 2022; NEDA, 2023).

To address these challenges, JeepniGo is proposed as a digital platform designed to enhance the traditional jeepney experience. By integrating real-time tracking, digital booking, cashless payments, and PWD-friendly options, JeepniGo aims to provide a seamless and accessible transport experience for both passengers and drivers. This initiative aligns with the PUVMP, supporting drivers through digital solutions and improving the overall commuter experience. Ultimately, JeepniGo aims to provide an additional income stream for drivers and resolve the daily difficulties faced by commuters, especially PWDs, ensuring the jeepney remains a viable and inclusive mode of transport in the modern era.

Purpose and Description

The JeepniGo capstone project aims to address the operational inefficiencies of the jeepney transportation system through a technological solution. By introducing a digital booking and fare collection platform, JeepniGo seeks to enhance the commuter experience and optimize driver operations. The app integrates ride-hailing, digital payments, and real-time tracking to ensure safer, more efficient, and transparent jeepney services in the Philippines.

1. Rationale, Timeliness, and Relevance: JeepniGo emerges as a response to the long-standing inefficiencies in the jeepney transportation system, such as unclear fare collection, lack of route optimization, and operational challenges that impact both drivers and passengers. Despite the jeepney's role as a primary mode of transportation for millions of Filipinos, the system remains largely manual, disorganized, and prone to delays. The increasing demand for convenient, transparent, and modernized public transport solutions underscores the need for JeepniGo.

By leveraging digital technology, JeepniGo aims to benefit commuters who seek reliable transport, drivers who struggle with unpredictable income and route inefficiencies, and operators who require better fleet management. Additionally, the project aligns with the Philippine government's Public Utility Vehicle Modernization Program (PUVMP), supporting efforts to digitize and improve the transportation sector.

2. Possible Solutions and Improvements: JeepniGo proposes a comprehensive digital platform that transforms jeepney operations by addressing common problems such as cash-based transactions, inefficient dispatching, and passenger uncertainty. The system introduces:

- Digital fare collection to enhance security and transparency.
- Real-time tracking to inform passengers of jeepney locations and minimize long waiting times.
- Route optimization to improve fuel efficiency and service reliability.
- Passenger booking system to reduce overcrowding and improve ride scheduling

3. Beneficiaries and Their Benefits: JeepniGo aims to benefit multiple stakeholders in the transportation sector:

- Daily commuters including PWD's - Enjoy faster, more predictable, and safer travel experiences. This includes all commuters benefiting from real-time tracking, digital booking, and digital payments. Furthermore, PWD commuters will experience enhanced accessibility through features like real-time tracking of PWD-friendly jeepneys, pre-booking options to ensure accessible seating, and driver notifications for assisted boarding.
- PUJ Drivers - Gain consistent income, reduced cash handling risks, and better route planning.
- Operators - Improve fleet monitoring and revenue management.

- Government Agencies - Support efforts in modernizing public transport and enhancing data-driven policy-making.

4. Contribution to Knowledge: The project presents a modernized approach to traditional jeepney operations, filling a gap in transport technology adoption in the Philippines. While ride-hailing and digital fare collection systems are widely used in other countries, their application to public jeepneys remains underdeveloped. JeepniGo contributes to the growing body of research on how technology can improve mass transportation, particularly in emerging economies.

The study highlights the importance of integrating digital solutions into public transport and serves as a reference for future innovations in the transportation industry.

5. Possible Implications: The successful implementation of JeepniGo could pave the way for broader transportation reforms in the Philippines. By demonstrating how technology-driven solutions can increase efficiency and commuter satisfaction, this project may:

- Influence future policies on digital fare collection and smart transport systems
- Encourage investment in transport technology and infrastructure
- Inspire similar innovations in other modes of public transport, such as buses, tricycles, and ferries

In practical terms, JeepniGo could revolutionize how jeepney operators run their businesses, making the system more competitive, sustainable, and commuter-friendly. Ultimately, the project aims to enhance daily travel experiences, improve drivers' livelihoods, and support the country's push toward a modernized transport sector.

Problem Statement and Objective

General Problem

- Limited information on passengers' demand.

Specific Problem

- Lack of clear communication for pick-up and drop-off location.
- Inefficient fare collection from passengers.

General Objective

- Increase passenger demand visibility to at least 80% during peak hours.

Specific objective

- Reduce the average waiting time for passengers to secure a seat in a jeepney to 30 minutes or less.
- Increase fare compliance rate among passengers to at least 95% through digital tracking and automated fare collection.

Capstone Project Gap Objectives

JeepniGo aims to bridge critical gaps in the jeepney sector by introducing modern digital solutions to improve efficiency, transparency, and overall commuter experience.

Efficient Booking and Fare Collection

JeepniGo introduces a streamlined digital booking and payment system, allowing passengers to reserve seats in advance and pay fares digitally. This innovation eliminates fare disputes between passengers and drivers, reduces long waiting times at terminals, and ensures a consistent fare collection process that improves financial stability for drivers.

Real-Time Jeepney Tracking

By implementing GPS-powered tracking, JeepniGo enables passengers to monitor the location of available jeepneys in real time. This feature minimizes uncertainty about jeepney arrival times, prevents overcrowding at stops due to lack of route visibility, and reduces delays caused by inefficient route planning.

Route Optimization for Efficiency

JeepniGo incorporates intelligent route-planning features to prevent overlapping routes and unnecessary detours. By leveraging data-driven insights, operators can optimize schedules based on passenger demand, reduce unnecessary fuel consumption and

operational costs, and ensure faster and more efficient travel times for commuters.

Digital Fare Transparency and Security

With cashless transactions, JeepniGo enhances security for both drivers and passengers by reducing the risks of theft and fare disputes. A standardized fare system ensures that passengers pay the correct amount without overcharging, enables drivers and operators to track earnings efficiently, and provides financial security through digital record-keeping.

Scope and Limitation of the Study

JeepniGo is specifically designed for jeepneys operating in Davao City, addressing challenges unique to the local public transportation system. While other cities and municipalities in the Philippines have their own systems, JeepniGo focuses on modernizing Davao City's jeepney network through digital solutions that enhance efficiency, security, and commuter convenience.

Key Features:

- Digital Booking System - Enables passengers to reserve seats in advance, reducing waiting times and overcrowding.
- Cashless Payment Integration - Supports secure transactions through mobile wallets and digital payments, reducing reliance on cash.

- Real-Time GPS Tracking - Allows commuters to monitor jeepney locations, estimated arrival times, and route availability.
- Route Optimization - Uses data analytics to streamline routes, minimize detours, and reduce fuel costs.
- Driver and Operator Dashboard - Provides a management system for tracking earnings, trip history, and schedules.
- Accessibility Features - Ensures an inclusive commuting experience by integrating features for persons with disabilities (PWDs).

Limitation of the Study

While JeepniGo offers promising solutions, certain constraints must be considered:

- Limited to registered jeepney operators - Only operators compliant with government regulations can integrate the system.
- Dependent on smartphone and internet access - Both drivers and passengers need compatible devices and stable internet connections to use the platform effectively.
- Initial implementation in select areas - The project will first focus on major urban centers, such as Davao City, before expanding nationwide.
- Potential cost for premium features - While basic services will be free, some advanced features (e.g., priority booking,

additional tracking features) may come with a subscription fee.

- Relies on government cooperation - Successful implementation requires collaboration with transport agencies and regulatory bodies.

With an estimated 2.3 million daily jeepney passengers in Davao City, JeepniGo has the potential to revolutionize the commuting experience, making it more reliable, efficient, and modernized.